

Edit Propagation, Representation Geometry, and Localization in Decoder-Only Knowledge Editing

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Background: decoder-only knowledge editing, locate-and-edit methods, and evaluation goals

Knowledge editing aims to change a small piece of a decoder-only LM's stored factual behavior without full retraining. In GPT-style models, a major line of work argues that some factual associations are recalled through localized middle-layer computations, which makes direct parameter edits possible and motivates evaluation in terms of edit success, side effects, and transfer to related prompts. (6 sources)

Knowledge editing is motivated by a simple deployment problem: pretrained language models can contain stale, wrong, or undesirable facts, but retraining or broad finetuning is costly and can change much more than the target behavior. Recent work therefore treats model editing as a way to directly change stored facts inside a model's parameters, often with no extra training at edit time. In decoder-only transformers, a central family is **locate-and-edit** methods, which first identify where a fact is processed or stored and then update only those parameters or activations most tied to that fact. This family includes **ROME**, **MEMIT**, and later extensions such as **EMMET**, and has become a standard framing for factual editing in GPT-style models. ([Yoon et al., 2024](#)) ([Gupta et al., 2025](#))

The key background claim behind this line of work is that factual associations in autoregressive transformers are not uniformly spread across the whole network in a way that is impossible to target. Instead, early influential evidence from GPT-style models suggested that factual recall depends on **localized, directly editable computations**, especially in **middle-layer feed-forward/MLP blocks** when the model processes the subject token. Meng et al. introduced causal tracing to identify activations that are decisive for factual prediction and then used that localization to build **ROME**, a rank-one weight update that changes one factual association at a time while trying to keep other behavior intact. They reported strong results on zero-shot relation extraction and on a counterfactual benchmark designed to test both specificity and generalization. This paper is the notable starting point for decoder-only factual editing. ([Meng et al., 2022](#)) ([Yoon et al., 2024](#)) ([Zhang et al., 2025](#))

From this base, the field split into a few broad editing styles. One branch modifies model weights directly, as in ROME and the batch-editing method **MEMIT**, which extends the same basic idea to many facts at once. Another branch learns editors or auxiliary mechanisms, such as **MEND**, which uses learned gradient-based

edits, or memory/external-cache approaches such as **GRACE** and **SERAC** that aim to support many updates with limited interference. For the purposes of this review, the important distinction is that locate-and-edit methods are especially tied to claims about **where factual knowledge lives inside decoder-only transformers**, because they depend on localizing and changing those internal computations rather than relying only on learned post hoc patching. ([Yoon et al., 2024](#)) ([Gupta et al., 2025](#)) ([Zhang et al., 2024](#)) ([Hartvigsen et al., 2022](#)) ([Zhang et al., 2025](#)) (Model-Generated)

A useful unifying view is that these locate-and-edit methods can be seen as optimizing a shared tradeoff between changing the target fact and preserving unrelated model behavior. Yoon et al. summarize later work showing that ROME, MEMIT, and related methods can be cast under a common **preservation-memorization** objective: write the new fact while preserving outputs for cached hidden states drawn from broad background data. Gupta et al. likewise note that ROME, MEMIT, and EMMET all include a precomputation stage that caches intermediate representations from many Wikipedia examples, and that the edit objective tries to preserve those outputs during the parameter update. This is important background because it explains why edit methods are always judged not just by whether they flip the target answer, but also by whether they avoid broader drift. ([Yoon et al., 2024](#)) ([Gupta et al., 2025](#))

As a result, evaluation in decoder-only model editing has converged on a few core goals. A successful edit must first have **reliability**: after the edit, the model should give the desired new answer for the edited prompt. It should also have **locality** or **specificity**: unrelated prompts should stay the same. And because factual changes should often affect nearby prompts and paraphrases, many papers also evaluate **portability** or robust generalization, including whether the edit transfers to subject substitutions, one-hop implications, or reverse relations. Zhang et al. make this three-part framing explicit and use it to compare editing methods, which is useful here because the user's question is ultimately about when edits propagate beyond the exact edited prompt. ([Zhang et al., 2024](#))

This background also clarifies the central tension that motivates the rest of the literature reviewed below. If facts are truly localized enough to be edited cleanly, then changing one fact might stay narrow and predictable. But if factual behavior depends on distributed or relational structure, then an edit can fail to generalize when it should, or propagate in unwanted ways when nearby facts share internal machinery. The whole modern literature on ripple effects, edit transfer, representational geometry, and mechanistic localization can be read as trying to understand exactly where decoder-only models fall on that spectrum in GPT-2, GPT-J, and later LLaMA-family systems. ([Meng et al., 2022](#)) ([Yoon et al., 2024](#)) ([Zhang et al., 2024](#)) ([Zhang et al., 2025](#))

Evidence

(Yoon et al., 2024)

[Is Bigger Edit Batch Size Always Better? - An Empirical Study on Model Editing with Llama-3](#)

"Traditional approaches often require retraining models on extensive datasets, a process that is both time-consuming and resourceintensive. An alternative is model editing (Yao et al., 2023), which allows for the modification of stored facts within a model, as well as the correction of inaccuracies. Several popular methods have emerged that infuse knowledge into models without the need for an additional hypernetwork, such as ROME (Rank-One Model Editing) (Meng et al., 2022), MEMIT (Mass Editing Memory in Transformer) (Meng et al., 2022b), and EMMET (Equality-constrained Mass Model Editing algorithm for Transformers) (Gupta et al., 2024b). These methods, traditionally called "locate-and-edit" algorithms, were recently shown to optimize the same objective, known as the preservation-memorization (PM) objective (Gupta et al., 2024b). They directly modify specific "knowledge-containing" areas of the model without necessitating additional training, and are applicable to any transformer-based large language models (LLMs)."

(Gupta et al., 2025)

[Efficient Knowledge Editing via Minimal Precomputation](#)

"Knowledge editing (Yao et al., 2023), or the ability to edit knowledge stored within the parameters of large language models (LLMs), is a topic of growing interest. A specific type of parameter-modifying knowledge editing methods called "locate-then-edit" methods (Yao et al., 2023) allow us to edit any transformer-based LLMs without the need for additional training. The most popular of these methods are MEMIT (Meng et al., 2022b), the first successful method that allows for batched editing, its predecessor ROME (Meng

et al., 2022a; Gupta et al., 2024a), which allows only one edit at a time, and EMMET (Gupta et al., 2024c), that generalizes ROME to batched editing"

". Each of MEMIT, ROME, and EMMET has a precomputation step where a large number of Wikipedia articles are passed through the model being edited and the intermediate hidden representations of the edited layers are cached. The editing loss function aims to preserve the outputs of these cached hidden representations during the editing process (Gupta et al., 2024c)."

(Meng et al., 2022)

Locating and Editing Factual Associations in GPT

"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at <https://rome.baulab.info/>"

(Zhang et al., 2025)

Memory in Large Language Models: Mechanisms, Evaluation and Evolution

"Under a unified operational definition, we define LLM memory as a persistent state written during pretraining, finetuning, or inference that can later be addressed and that stably influences outputs"

".For updating and forgetting, we present DMM Gov: coordinating DAPT/TAPT, PEFT, model editing (ROME, MEND, MEMIT, SERAC), and RAG to form an auditable loop covering admission thresholds, rollout, monitoring, rollback, and change audits, with specs for timeliness, conflict handling, and long-horizon consistency"

".The paper Locating and Editing Factual Associations in GPT was the first to prove that factual associations in autoregressive transformer models (such as GPT) are stored in locally editable computational modules, rather than being scattered across the entire network. This provides a theoretical basis for "precise editing" of parametric memory [145]. As stated in the paper: "We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations.""

"Core Methodology: Causal Tracing and ROME Editing -Causal Tracing: This method locates knowledge storage modules through causal mediation analysis, involving three model runs: 1. Clean run: Input a complete factual prompt (e.g., "Paris is the capital of [Mask]") and record the model's internal hidden states; 2. Corrupted run: Replace the topic word with an irrelevant word (e.g., "Xyz is the capital of [Mask]") and observe the model's prediction bias; 3. Corrupted-with-restoration run: During the corrupted run, restore only the hidden state of a specific module in a specific layer to the result from the clean run. If the model's prediction accuracy recovers, this module is identified as critical for storing the fact"

".The Average Indirect Effect (AIE) is calculated to quantify the module's contribution-the higher the AIE, the stronger the causal impact of the module on fact recall"

".-ROME (Rank-One Model Editing): Based on the results of causal tracing, a "rank-one update" method is proposed to modify facts. For middle-layer MLP modules (which causal tracing identified as having the highest AIE), the weight matrix is updated with a rank-one matrix to precisely replace a specific fact (e.g., changing "Paris is the capital of France" to "Paris is the capital of Japan") without modifying other parameters."

(Zhang et al., 2024)

OneEdit: A Neural-Symbolic Collaboratively Knowledge Editing System

"The baselines include methods for continual learning and model editing. We compare OneEdit against direct fine-tuning (FT) with an additional KL divergence loss, as well as ROME, MEMIT, and GRACE (Hartvigsen et al., 2022)(Meng et al., 2022)[29]"

".To evaluate the effectiveness of an editing method, we primarily consider three aspects: Reliability, Locality, and Portability. Reliability, as defined by Huang et al. [19], refers to a successful edit when the post-edit model M' provides the edited target answer ' for the prompt"

".Locality, also referred to as Specificity in some works, denotes that editing should be implemented locally. This means that the post-edit model M' should not alter the output of irrelevant examples in the out-of-scope set $(,)$ "

"Portability, as defined by Yao et al. [55], encompasses three components: Subject-Replace, One-hop, and Reverse. Portability is used to comprehensively evaluate the effectiveness of model editing in transferring knowledge to related content, termed robust generalization."

(Hartvigsen et al., 2022)

[Aging with GRACE: Lifelong Model Editing with Discrete Key-Value Adaptors](#)

"Deployed language models decay over time due to shifting inputs, changing user needs, or emergent world-knowledge gaps. When such problems are identified, we want to make targeted edits while avoiding expensive retraining. However, current model editors, which modify such behaviors of pre-trained models, degrade model performance quickly across multiple, sequential edits. We propose GRACE, a lifelong model editing method, which implements spot-fixes on streaming errors of a deployed model, ensuring minimal impact on unrelated inputs. GRACE writes new mappings into a pre-trained model's latent space, creating a discrete, local codebook of edits without altering model weights. This is the first method enabling thousands of sequential edits using only streaming errors. Our experiments on T5, BERT, and GPT models show GRACE's state-of-the-art performance in making and retaining edits, while generalizing to unseen inputs. Our code is available at <https://www.github.com/thartvigsen/grace>."

Knowledge editing methods in decoder-only LMs: ROME, MEMIT, MEND, and why edits fail to generalize or scale

In decoder-only models, ROME, MEMIT, and MEND differ mainly in how they compute and apply an edit: direct low-rank weight change to localized MLP layers, multi-layer batch insertion of many memories, or learned gradient-to-update mapping. Across GPT-2, GPT-J, and later LLaMA-style studies, the main failure modes are not just missed target edits, but poor semantic scope control, interference across edits, forgetting under sequential updates, and worse behavior when the new fact contradicts strongly connected old knowledge. (17 sources)

Within decoder-only language models, the notable split is between **direct weight editing** and **learned editors**. **ROME** is the clearest direct editing method: it uses causal mediation/tracing to find the middle-layer feed-forward modules that matter for recalling a subject's fact, then applies a **rank-one update** to the selected MLP weights so the model maps the subject to a new value. Meng et al. argue that factual associations are recalled through localized, directly editable computations in mid-layer MLPs, and show that ROME can change a single fact while keeping good specificity and generalization on CounterFact better than prior alternatives that tended to trade one off against the other. Later summaries consistently describe ROME this way: as a locate-then-edit method that edits a single factual association by modifying feed-forward weights in the causally important middle layers of GPT-style models. ([Meng et al., 2022](#)) ([Mazzia et al., 2023](#)) ([Bi et al., 2024](#)) ([Qin et al., 2024](#)) ([Wang et al., 2025](#)) ([Sun et al., 2025](#))

MEMIT extends the same basic view of factual storage, but changes the scaling story. Instead of assuming one fact can be rewritten through one layer, MEMIT spreads the update across a **sequence of causal mediator layers** and is designed to perform **many edits at once**, with reports of scaling to thousands of associations in GPT-J and GPT-NeoX. Meng et al. frame it as direct memory insertion into transformer weights while maintaining efficacy, generalization, and specificity; later reviews describe MEMIT as the main batch-editing extension of ROME for decoder-only transformers. This is why MEMIT is often treated as the default locate-and-edit baseline when the question is scalability rather than one-shot precision. ([Meng et al., 2022](#)) ([Gupta et al., 2024](#)) ([Cao et al., 2025](#)) ([Wang et al., 2025](#)) ([Sun et al., 2025](#)) ([Xu et al., 2025](#))

By contrast, **MEND** does not directly solve a per-edit linear memory update in the model weights. It is a **meta-learned hypernetwork** editor that takes the gradient of the desired edit and predicts a low-rank weight update, with the goal of making local changes efficiently on large pretrained models. In later decoder-only comparisons, MEND is usually a baseline against ROME and MEMIT rather than a localization-driven method in the same mechanistic sense. This difference matters for the user's question: MEND can succeed behaviorally without committing as strongly to the claim that one fact sits in one geometrically neat internal locus. ([Meng et al., 2022](#)) ([Gupta et al., 2024](#)) ([Bi et al., 2024](#)) ([Qin et al., 2024](#))

A key reason direct editing looked plausible in the first place is the broader **MLP-as-memory** picture. Prior analysis of transformer feed-forward layers described them as **key-value memories**, where keys match input patterns and values push the next-token distribution toward associated concepts. This line of work, along

with “knowledge neuron” studies, gave the field a concrete hypothesis: if factual recall uses MLP key-value structure, then one can sometimes rewrite a fact by changing the relevant mapping rather than fine-tuning the whole model. ROME and MEMIT are best understood as operationalizing that hypothesis for decoder-only factual editing. ([Gupta et al., 2023](#)) ([Geva et al., 2020](#)) ([Geva et al., 2022](#)) ([Dai et al., 2021](#)) ([Clemente et al., 2025](#))

The harder question is **why these edits often fail to generalize cleanly**. One failure mode is **insufficient scope modeling**: editing methods may flip the exact edited prompt but miss paraphrases, nearby prompts, or logically related cases, because the method does not accurately capture the full set of inputs that should change. This criticism was made clearly for earlier learned editors and motivates systems like SERAC, whose authors argue that existing editors struggle to represent an edit’s intended scope and therefore make wrong predictions on loosely related inputs. The same issue appears in factual and commonsense direct editing work, which emphasizes that success on the original benchmark prompt is not enough; one must also test affected paraphrases, affected neighborhoods, and reasoning chains. ([Mitchell et al., 2022](#)) ([Gupta et al., 2023](#))

A second failure mode is **interference as edits accumulate**. Even when single-edit results are strong, sequential or large-scale editing can make the model forget prior edits and degrade downstream behavior. Gupta et al. report that in GPT2-XL and GPT-J, sequential editing with ROME or MEMIT leads to a two-phase pattern: first gradual forgetting, then abrupt or catastrophic forgetting. A 2025 analysis similarly describes long-run decline in editing success as **superimposed noise accumulation**, tied to erroneous activation of irrelevant knowledge and conflicts among activated knowledge. These findings are directly relevant to the user’s propagation question: once multiple nearby facts compete in overlapping circuitry, clean isolation becomes harder, so both under-propagation and over-propagation become more likely. ([Gupta et al., 2024](#)) ([Cao et al., 2025](#))

A third and deeper failure mode is **contradiction sensitivity**. Later work argues that LLMs can often absorb non-conflicting new facts with limited damage, but **contradictory updates** trigger much stronger interference with unrelated knowledge. Clemente et al. report severe corruption of unrelated knowledge when models are forced to learn contradictory facts, and they find that even targeted updates to supposedly more “plastic” neurons do not remove this problem across scales from GPT-2 to GPT-J. This suggests that some failures are not just due to bad layer choice or poor optimization, but to the fact that the edited association is entangled with a broader network of already-used representations. ([Clemente et al., 2025](#))

The practical outcome is a nuanced ranking of methods rather than a single winner. **ROME** remains the notable single-fact mechanistic editor for decoder-only models because it ties editability to causal localization and gives clean one-shot interventions. **MEMIT** is the notable scaling extension because it can batch many edits across layers and often outperforms alternatives when many updates are needed, including in later GPT-2/GPT-J/LLaMA comparisons. **MEND** remains important as a flexible learned baseline, but the literature repeatedly finds that learned or direct editors alike run into scope, interference, and forgetting problems once one moves beyond isolated benchmark edits. ([Meng et al., 2022](#)) ([Meng et al._1, 2022](#)) ([Gupta et al., 2024](#)) ([Shafqat et al., 2024](#))

For the user’s core question, this section sets up the main answer: current decoder-only editing methods are built on the idea that some facts are locally writable, but their failures show that **successful editing is not determined only by hitting the right layer**. Generalization and propagation depend on how the target fact sits inside a larger representational and causal network—shared MLP memories, paraphrase scope, neighboring entities, contradictory relations, and interference from prior edits—which is exactly why later work turns to ripple-effect benchmarks, geometry, and causal localization tools. ([Meng et al., 2022](#)) ([Gupta et al., 2023](#)) ([Gupta et al., 2024](#))

Evidence

(Meng et al., 2022)

Locating and Editing Factual Associations in GPT

"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at <https://rome.baulab.info/>"

(Mazzia et al., 2023)

A Survey on Knowledge Editing of Neural Networks

"Firstly, they trace the causal effects of hidden state activations within GPT using causal mediation analysis (Pearl, 2001) as previously done by other works (Vig et al., 2020) to identify the specific modules that mediate recall of a fact about a subject. Their analysis reveals that feedforward MLPs at a range of middle layers are decisive when processing the last token of the subject name. Secondly, they test their previous findings by introducing a novel direct model editing methodology: Rank-One Model Editing method (ROME). The algorithm aims at producing single, not successive edits by altering the parameters that determine a feedforward layer's behavior. As stated by the authors, despite the simplicity of the intervention, ROME is similarly effective to other model-editing approaches achieving good generalization and locality simultaneously, whereas previous approaches sacrifice one or the other. Furthermore, they introduce a new dataset, CounterFact (derived from a set of true facts from WikiData), for evaluating counterfactual edits in language models."

(Bi et al., 2024)

StruEdit: Structured Outputs Enable the Fast and Accurate Knowledge Editing for Large Language Models

"ROME (Meng et al., 2022) applies causal mediation analysis to locate the editing area, framing model editing as a least-squares problem under linear equality constraints and solving it using lagrange multipliers"

".MEND (Mitchell et al., 2021) adopt a meta-learning approach that trains a hypernetwork to infer weight updates from the gradient of the inserted fact"

".MEMIT (Meng et al., 2022b) insert new memories into language models by targeting key transformer weights identified as causal mediators of factual knowledge recall."

(Qin et al., 2024)

Mitigating Gender Bias in Code Large Language Models via Model Editing

"Meng et al. (Meng et al., 2022) propose a method called Rank-One Model Editing (ROME), which updates specific factual associations by directly modifying the weights of the feedforward layers, thereby achieving precise editing of the model while keeping other parts of the model unaffected. Mitchell et al. [45] propose an efficient model editing method called Model Editor Networks with Gradient Decomposition (MEND), which rapidly and locally modifies the behavior of large pre-trained models by using small auxiliary editing networks and gradient decomposition."

(Wang et al., 2025)

MEMOIR: Lifelong Model Editing with Minimal Overwrite and Informed Retention for LLMs

"A.3 Baselines ROME (Meng et al., 2022) identified the feedforward MLPs in the middle layers store the factual knowledge by tracing the causal effects of hidden state activations based on causal mediation analysis. Furthermore, they introduced a Rank-One Model Editing method to alter the parameters storing factual knowledge and modify the model's behavior."

MEMIT [14] extends ROME to batch edits and multiple layers."

(Sun et al., 2025)

Mitigating Negative Interference in Multilingual Sequential Knowledge Editing through Null-Space Constraints

"ROME (Rank-One Model Editing) (Meng et al., 2022) localizes factual associations to middle-layer feed-forward modules and updates their weights through rank-one adjustments. By identifying critical neuron activations tied to specific knowledge, ROME demonstrates that direct manipulation of feed-forward layers can edit factual predictions without requiring full retraining. MEMIT (Mass-Editing Memory in Transformer) (Meng et al., 2023) extends ROME by propagating edits across multiple layers. It identifies critical neuron activations in parallelized weight matrices during knowledge editing, enabling the insertion of thousands of new

associations"

"Prior research (Geva et al., 2021) has demonstrated that the MLP layers in Transformer models function as knowledge repositories, with specific neurons encoding editable factual associations.

Building on this insight, Meng et al. (2022) introduced ROME (Rank-One Model Editing), a pioneering two-stage framework that first identifies the MLP layer storing the target knowledge and then injects a single knowledge edit through rankone weight perturbations. This approach was later extended by MEMIT (Mass-Editing Memory in Transformer) (Meng et al., 2023), which enhances scalability by enabling updates across multiple MLP layers to inject numerous knowledge edits."

(Meng et al., 2022)

Mass-Editing Memory in a Transformer

"De (Cao et al., 2021) proposes a Knowledge Editor (KE) hypernetwork that edits a standard model by predicting updates conditioned on new factual statements. In a study of KE, Hase et al. (2021) find that it fails to scale beyond a few edits, and they scale an improved objective to 10 beliefs. MEND (Mitchell et al., 2021) also adopts meta-learning, inferring weight updates from the gradient of the inserted fact. To scale their method, (Mitchell et al., 2022) proposes SERAC, a system that routes rewritten facts through a different set of parameters while keeping the original weights unmodified; they demonstrate scaling up to 75 edits. Rather than meta-learning, our method employs direct parameter updates based on an explicitly computed mapping"

"Our method is closely related to (Meng et al., 2022), which also updates GPT as an explicit associative memory. Unlike the single-edit approach taken in that work, we modify a sequence of layers and develop a way for thousands of modifications to be performed simultaneously"

"MEMIT targets the weights of transformer modules that we determine to be causal mediators of factual knowledge recall"

"In practical settings, we may wish to MEMIT modifies transformer weights to edit memories, e.g., "Michael Jordan now plays the sport baseball," while (c) maintaining generalization, specificity, and fluency at scales beyond other methods. As Section 5.2.2 details, editing score is the harmonic mean of efficacy, generalization, and specificity metrics."

(Gupta et al., 2024)

Model Editing at Scale leads to Gradual and Catastrophic Forgetting

"In this paper, we focus on two prominent model editing methods in literature -Rank-One Memory Editing (ROME) (Meng et al., 2022) and Mass-Editing Memory in a Transformer (MEMIT) (Meng et al., 2022b), whereas Model Editor Networks using Gradient Decomposition (MEND) (Mitchell et al., 2021) and fine-tuning are used as baselines. MEND is a hypernetwork (Chauhan et al., 2023) based model editing method that generates the weight updates for the model being edited. ROME and MEMIT first localize knowledge within a model using causal tracing (Vig et al., 2020) and then update the weights of the selected layers to inject knowledge. The major difference between ROME and MEMIT is that while ROME works under the assumption that knowledge in LLMs can be updated by updating a single layer, MEMIT updates the weights of multiple layers. We evaluate these model editing methods over two models -GPT2-XL (1.5B) (Radford et al., 2019) and GPT-J (6B) (Wang and Komatsuzaki, 2021). Note that all prior works (Mitchell et al., 2021; Meng et al., 2022a,b) edit base language models and not chat models"

"We find that as the model is edited sequentially with multiple facts, it continually forgets previously edited facts and the ability to perform downstream tasks. This forgetting happens in two phases -- an initial gradual but progressive forgetting phase followed by abrupt or catastrophic forgetting phase. Both gradual and catastrophic forgetting limit the usefulness of model editing methods at scale -- the former making model editing less effective as multiple edits are made to the model while the latter caps the scalability of such model editing methods. Our analysis also highlights other key limitations of ROME and MEMIT at scale."

(Cao et al., 2025)

On the Superimposed Noise Accumulation Problem in Sequential Knowledge Editing of Large Language Models

"ROME: ROME is a method designed to update specific factual associations in large language models. By identifying key neuron activations in the feed-forward networks of middle layers that influence factual predictions, ROME directly modifies the weights of feedforward layers to edit these associations. ROME demonstrates that the feedforward modules in intermediate layers play a critical role in storing and recalling factual knowledge, making direct manipulation of the model a feasible editing technique"

"MEMIT: MEMIT is a scalable multilayer update algorithm designed to efficiently insert new factual memories into transformer-based language models. Building on the direct editing approach of ROME, MEMIT targets specific weights in transformer modules that act as causal mediators for factual knowledge recall. This approach allows MEMIT to update thousands of new factual associations within the model"

"Sequential knowledge editing techniques aim to continuously update knowledge in large language models at low cost, preventing models from generating outdated or incorrect information. However, existing sequential editing methods suffer from a significant decline in editing success rates after long-term editing. Through theoretical analysis and experiments, our findings reveal that as the

number of edits increases, the model's output increasingly deviates from the desired target, leading to a drop in editing success rates. We refer to this issue as the superimposed noise accumulation problem. Our further analysis demonstrates that the problem is related to the erroneous activation of irrelevant knowledge and conflicts between activated knowledge."

(Xu et al., 2025)

MedMKEB: A Comprehensive Knowledge Editing Benchmark for Medical Multimodal Large Language Models

"MALMEN (Tan, Zhang, and Fu 2024) transforms the parameter offset aggregation problem required for editing into a least squares problem, and updates the model parameters by solving normal equations to ensure accuracy and stability during large-scale editing. At the same time, the algorithm separates the calculation process of the hypernetwork and the basic language model, allowing both to support operations of arbitrary batch sizes under limited memory, breaking through the scale limit of simultaneous editing"

".Locate-then-Edit These approaches first locate the target knowledge in model parameters or activations and then apply localized updates. Knowledge Neurons (KN) (Dai et al., 2021) uses BERT as the research object, designs a knowledge attribution method for cloze tasks, and identifies neurons that are strongly related to specific facts. Experiments show that the activation level of such neurons is positively correlated with the accuracy of the expression of the corresponding facts, proving that they are the key carriers of factual knowledge. Based on the positioning results of knowledge neurons, attempts are made to directly edit the target knowledge by adjusting the state of specific knowledge neurons (such as updating or erasing) without full fine-tuning. Although this exploration is still in its early stages, it provides new ideas for "precisely modifying model knowledge and multiple facts with strong locality and generalization guarantees". MEMIT (Meng et al., 2022)) is a method that supports large-scale memory editing of Transformer models. This method breaks through the previous limitations of editing a single or a small amount of knowledge. The core goal is to achieve "batch knowledge update": by optimizing the model parameter adjustment strategy, the pre-trained language model can simultaneously receive and store a large number of new memories without the need to fully retrain the model."

(Gupta et al., 2023)

Editing Common Sense in Transformers

"Inspired by the linear associative memory property of feedforward layers in Transformers (Anderson, 1972)(Geva et al., 2020)(Geva et al., 2022) and success with the approach in convolutional models (Bau et al., 2020), recent works have proposed to edit MLP weights directly (Meng et al., 2022)(Dai et al., 2021)(Yao et al., 2022). In the encyclopedic factual domain, (Meng et al., 2022) proposed to edit single facts by fitting a Rank One Model Edit (ROME) to the parameters of an MLP layer, and showed it outperformed prior methods. Our work builds on Meng et al. (2023), which extended this approach to thousands of edits by altering the weights of a range of MLP layers. Hase et al. (2023a) demonstrate that many early edit layers can work well with MEMIT; this partially motivates our extensive layer hyperparameter search. Recent work by Cohen et al. (2023) proposes a dataset for evaluation of a variety of ripple effects in editing methods with factual knowledge and concludes that models fail to capture these effects"

".It is not enough to report the success of a direct editing method on the original dataset since edit methods can (and should) have propagational effects on instances beyond the dataset (Meng et al., 2022). To compare and assess semantic generalization of updates, we augment incorrectly predicted samples with neighborhood instances and paraphrases that should be affected by an edit, similar to the prior fact editing work. We additionally include neighborhood instances that should not be affected. Performance on the unaffected neighborhood measures the update's specificity, while performance on the affected neighborhoods and affected paraphrases indicates its generalization. Additionally, editing the plausibility of a commonsense statement should affect reasoning chains involving that statement. Entities and knowledge are interconnected, often requiring updates to one component of commonsense knowledge when modifying another. To this end, we add a fourth category of augmentations, affected reasoning, to test whether successful edits correct aspects of a model's commonsense reasoning."

(Geva et al., 2020)

Transformer Feed-Forward Layers Are Key-Value Memories

"Feed-forward layers constitute two-thirds of a transformer model's parameters, yet their role in the network remains under-explored. We show that feed-forward layers in transformer-based language models operate as key-value memories, where each key correlates with textual patterns in the training examples, and each value induces a distribution over the output vocabulary. Our experiments show that the learned patterns are human-interpretable, and that lower layers tend to capture shallow patterns, while upper layers learn more semantic ones. The values complement the keys' input patterns by inducing output distributions that concentrate probability mass on tokens likely to appear immediately after each pattern, particularly in the upper layers. Finally, we demonstrate that the output of a feed-forward layer is a composition of its memories, which is subsequently refined throughout the model's layers via residual connections to produce the final output distribution."

(Geva et al., 2022)

Transformer Feed-Forward Layers Build Predictions by Promoting Concepts in the Vocabulary Space

"Transformer-based language models (LMs) are at the core of modern NLP, but their internal prediction construction process is opaque and largely not understood. In this work, we make a substantial step towards unveiling this underlying prediction process, by reverse-engineering the operation of the feed-forward network (FFN) layers, one of the building blocks of transformer models. We

view the token representation as a changing distribution over the vocabulary, and the output from each FFN layer as an additive update to that distribution. Then, we analyze the FFN updates in the vocabulary space, showing that each update can be decomposed to sub-updates corresponding to single FFN parameter vectors, each promoting concepts that are often human-interpretable. We then leverage these findings for controlling LM predictions, where we reduce the toxicity of GPT2 by almost 50%, and for improving computation efficiency with a simple early exit rule, saving 20% of computation on average."

(Dai et al., 2021)

Knowledge Neurons in Pretrained Transformers

"Large-scale pretrained language models are surprisingly good at recalling factual knowledge presented in the training corpus. In this paper, we present preliminary studies on how factual knowledge is stored in pretrained Transformers by introducing the concept of knowledge neurons. Specifically, we examine the fill-in-the-blank cloze task for BERT. Given a relational fact, we propose a knowledge attribution method to identify the neurons that express the fact. We find that the activation of such knowledge neurons is positively correlated to the expression of their corresponding facts. In our case studies, we attempt to leverage knowledge neurons to edit (such as update, and erase) specific factual knowledge without fine-tuning. Our results shed light on understanding the storage of knowledge within pretrained Transformers."

(Clemente et al., 2025)

In Praise of Stubbornness: An Empirical Case for Cognitive-Dissonance Aware Continual Update of Knowledge in LLMs

"Next, a big portion of recent literature has focused on understanding and modifying the internal knowledge of Large Language Models (LLMs), post-training. Such knowledge editing aims to alter specific facts or associations within the model without the need for full retraining. (Geva et al., 2020) were among the first to show that transformer Feed-Forward Network (FFN) layers act as unnormalized key-value stores encoding relational knowledge inside LLMs. This observation was later confirmed and complemented by others (Meng et al., 2022), Dai et al., 2021] before being leveraged by subsequent work to master the editing of internal memories. (Meng et al., 2022) introduced ROME (Rank-One Model Editing), a method that uses causal tracing to empirically locate the layers essential to encoding a given association. They then modify these modules by applying small rank-one changes. To identify the relevant modules, they run the network multiple times, introducing corruptions to the input sequence to disturb the inference, and then restore individual states from the original non-corrupted pass. But this work and others worked only on single edits, and were often evaluated one edit at a time, starting each time from a fresh pre-trained model. The same authors later developed MEMIT, which follows the same causal tracing principle but with the goal of scaling up to 10,000 edits in bulk [Meng et al., 2022c]. Similarly, Dai et al."

".Through systematic empirical investigation, we uncover a fundamental and concerning property of Large Language Models: while they can safely learn facts that don't contradict their knowledge, attempting to update facts with contradictory information triggers catastrophic corruption of unrelated knowledge. Unlike humans, who naturally resist contradictory information, these models indiscriminately accept contradictions, leading to devastating interference, destroying up to 80% of unrelated knowledge even when learning as few as 10-100 contradicting facts. To understand whether this interference could be mitigated through selective plasticity, we experiment with targeted network updates, distinguishing between previously used (stubborn) and rarely used (plastic) neurons. We uncover another asymmetry: while sparing frequently-used neurons significantly improves retention of existing knowledge for non-contradictory updates (98% vs 93% with standard updates), contradictory updates trigger catastrophic interference regardless of targeting strategy. This effect which persists across tested model scales (GPT-2 to GPT-J-6B), suggests a fundamental limitation in how neural networks handle contradictions."

(Mitchell et al., 2022)

Memory-Based Model Editing at Scale

"Even the largest neural networks make errors, and once-correct predictions can become invalid as the world changes. Model editors make local updates to the behavior of base (pre-trained) models to inject updated knowledge or correct undesirable behaviors. Existing model editors have shown promise, but also suffer from insufficient expressiveness: they struggle to accurately model an edit's intended scope (examples affected by the edit), leading to inaccurate predictions for test inputs loosely related to the edit, and they often fail altogether after many edits. As a higher-capacity alternative, we propose Semi-Parametric Editing with a Retrieval-Augmented Counterfactual Model (SERAC), which stores edits in an explicit memory and learns to reason over them to modulate the base model's predictions as needed. To enable more rigorous evaluation of model editors, we introduce three challenging language model editing problems based on question answering, fact-checking, and dialogue generation. We find that only SERAC achieves high performance on all three problems, consistently outperforming existing approaches to model editing by a significant margin. Code, data, and additional project information will be made available at <https://sites.google.com/view/serac-editing>."

(Shafqat et al., 2024)

Evaluating Complex Entity Knowledge Propagation for Knowledge Editing in LLMs

"The study evaluates its proposed methodology using various evaluation metrics and four distinct editing methods across three diverse language models (GPT2-XL, GPT-J, and Llama-2-7b). The results indicate the superiority of mass-editing memory in a

transformer (MEMIT) and in-context learning for knowledge editing (IKE) in efficiently executing multiple updates within the triplet representation framework."

Ripple effects and multi-hop consequences of edits: RippleEdits, MQuAKE, and portability

Benchmarks after ROME and MEMIT showed that changing one fact rarely updates all the other facts that should follow from it. Models often learn the edited statement and even its paraphrases, but still fail on related one-hop and multi-hop questions, which means portability is much weaker than simple edit success suggests. (5 sources)

A major shift in the 2023 literature was the claim that standard edit evaluation was too narrow. Earlier benchmarks mostly asked whether the model now states the new target fact, but this misses the fact that one edited belief should often force changes in a larger web of related beliefs. Cohen et al. call these **ripple effects**: if one fact changes, other facts tied by family, role, or relation structure should also change, and they build **RippleEdits** to test exactly this kind of consistency over about 5K factual edits. Their main result is negative: prominent editing methods often do not introduce coherent downstream changes, even when the base edit itself succeeds. [\(Cohen et al., 2023\)](#)

Zhong et al. make the same point for **multi-hop QA** rather than local relational consistency. They argue that if a model is edited so that, for example, the UK Prime Minister is now Rishi Sunak, then answers to questions that depend on that fact should also change, such as questions about the British Prime Minister's spouse. To test this, they propose **MQuAKE**, a benchmark of multi-hop questions whose answers should update as entailed consequences of factual edits. Across methods including fine-tuning, MEND, ROME, and MEMIT, they find a striking gap: edited models can often recall the edited fact and answer paraphrases, but they "fail catastrophically" on the multi-hop consequences. In one example they report that a GPT-J model edited by ROME answers only 7.6% of MQuAKE-CF multi-hop questions correctly after the edit, despite doing much better on direct recall-style checks. [\(Zhong et al., 2023\)](#) [\(Mitchell et al., 2021\)](#) [\(Meng et al., 2022\)](#) [\(Zhu et al., 2020\)](#)

These benchmarks matter for the user's question because they separate **surface portability** from **reasoning-level propagation**. A model may update the exact edited prompt and semantically close paraphrases, yet still fail to carry that change through a chain of dependent facts. In other words, there is a big difference between "the edited fact is retrievable" and "the model's internal knowledge graph has been updated in a way that supports downstream inference." That gap is now one of the clearest empirical findings in decoder-only editing work on GPT-2, GPT-J, and related systems. [\(Zhong et al., 2023\)](#) [\(Cohen et al., 2023\)](#)

For portability evaluation, this leads to a more precise picture than the one used in early editing papers. There are at least three levels: direct recall of the edited fact, transfer to paraphrases or nearby prompts, and propagation to **related implied facts** or **multi-hop answers**. RippleEdits stresses the second-to-third step for local relational neighborhoods, while MQuAKE stresses it for compositional question answering over edited knowledge. Both papers show that current editors are much stronger at the first two levels than at the third. [\(Cohen et al., 2023\)](#) [\(Zhong et al., 2023\)](#)

This also sharpens why "did the edit work?" is the wrong question for many applications. If a user changes a biographical, geopolitical, or family fact, they usually want more than a one-sentence patch; they want the model to behave as if a connected set of consequences has changed. The failure cases in RippleEdits and MQuAKE therefore suggest that many decoder-only edits act more like **local response rewrites** than like deep belief revision, even when the underlying method is mechanistically grounded like ROME or scaled up like MEMIT. [\(Cohen et al., 2023\)](#) [\(Zhong et al., 2023\)](#) [\(Meng et al., 2022\)](#)

For the review's main theme, these results are also an indirect warning against a simple geometric story. If two facts are "close" in some intuitive semantic sense, current locate-and-edit methods still often do not update both in a consistent way. So any claim that representational similarity predicts edit propagation has to explain why edited models can preserve direct recall and paraphrase generalization while failing on one-hop

and multi-hop entailments. That is why later sections turn to internal geometry and causal localization: the key issue is not only whether two facts are related, but **how that relation is implemented inside the model's circuits and layers**. ([Zhong et al., 2023](#)) ([Cohen et al., 2023](#))

Evidence

(Cohen et al., 2023)

[Evaluating the Ripple Effects of Knowledge Editing in Language Models](#)

"Here we argue that such evaluation is limited, since injecting one fact (e.g., "Jack Depp is the son of Johnny Depp") introduces a "ripple effect" in the form of additional facts that the model needs to update (e.g., "Jack Depp is the sibling of Lily-Rose Depp"). To address this, we propose novel evaluation criteria that consider the implications of an edit on related facts. Using these criteria, we then construct RippleEdits, a diagnostic benchmark of 5K factual edits, capturing various types of ripple effects. We evaluate prominent editing methods on RippleEdits, showing that they fail to introduce consistent changes in the model's knowledge"

"We introduce the notion of ripple effects in knowledge editing, suggesting that editing a particular fact implies further updates of related facts. We additionally propose evaluation criteria for ripple effects and create RIPPLEEDITS, a diagnostic benchmark designed to evaluate how well KE methods handle the ripple effects of various edits. We evaluate prominent KE methods and show that they often fail to introduce consistent edits that capture the ripple effects of an edit, suggesting that future development of KE methods should consider those effects more carefully"

"Finally, it would be interesting to consider cases where models succeed in capturing ripple-edits, and analyze how these are implemented mechanistically in the transformer architecture (Geva et al., 2023)."

(Zhong et al., 2023)

[MQQuAKE: Assessing Knowledge Editing in Language Models via Multi-Hop Questions](#)

"Current evaluation paradigms are extremely limited, mainly validating the recall of edited facts, but changing one fact should cause rippling changes to the model's related beliefs. If we edit the UK Prime Minister to now be Rishi Sunak, then we should get a different answer to Who is married to the British Prime Minister? In this work, we present a benchmark, MQQuAKE (Multi-hop Question Answering for Knowledge Editing), comprising multi-hop questions that assess whether edited models correctly answer questions where the answer should change as an entailed consequence of edited facts. While we find that current knowledge-editing approaches can recall edited facts accurately, they fail catastrophically on the constructed multi-hop questions"

"We evaluate state-of-the-art knowledge-editing methods on MQQuAKE, testing from editing facts mentioned in one question to editing facts mentioned in a large set of questions. The latter setting evaluates approaches that are designed to handle many edits, such as MEMIT (Meng et al., 2022b). Surprisingly, existing knowledge-editing methods often perform well on answering questions that are paraphrases of the edited fact but fail drastically on questions where the answer should change as a consequence of an edited fact. For example, a GPT-J model edited by ROME (Meng et al., 2022) can only answer 7.6% of multi-hop questions in MQQuAKE-CF, even though it could answer 43.4% of the questions before editing"

"• Fine-tuning (FT) simply performs gradient descent on the edits to update model parameters. We follow (Zhu et al., 2020) and fine-tune one layer in the model with a norm constraint on weight changes"

"• MEND (Mitchell et al., 2021) trains a hypernetwork to produce weight updates by transforming the raw fine-tuning gradients given an edited fact"

"• ROME (Meng et al., 2022) first localizes the factual knowledge at a certain layer in the Transformer architecture, and then updates the feedforward network in that layer to insert the new facts"

"• MEMIT (Meng et al., 2022b)"

(Mitchell et al., 2021)

[Fast Model Editing at Scale](#)

"While large pre-trained models have enabled impressive results on a variety of downstream tasks, the largest existing models still make errors, and even accurate predictions may become outdated over time. Because detecting all such failures at training time is impossible, enabling both developers and end users of such models to correct inaccurate outputs while leaving the model otherwise intact is desirable. However, the distributed, black-box nature of the representations learned by large neural networks makes producing such targeted edits difficult. If presented with only a single problematic input and new desired output, fine-tuning approaches tend to overfit; other editing algorithms are either computationally infeasible or simply ineffective when applied to very large models. To enable easy post-hoc editing at scale, we propose Model Editor Networks using Gradient Decomposition (MEND), a collection of small auxiliary editing networks that use a single desired input-output pair to make fast, local edits to a pre-trained model's behavior. MEND learns to transform the gradient obtained by standard fine-tuning, using a low-rank decomposition of the

gradient to make the parameterization of this transformation tractable. MEND can be trained on a single GPU in less than a day even for 10 billion+ parameter models; once trained MEND enables rapid application of new edits to the pre-trained model. Our experiments with T5, GPT, BERT, and BART models show that MEND is the only approach to model editing that effectively edits the behavior of models with more than 10 billion parameters. Code and data available at <https://sites.google.com/view/mend-editing>."

(Meng et al., 2022)

Locating and Editing Factual Associations in GPT

"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at <https://rome.baulab.info/>"

(Zhu et al., 2020)

Modifying Memories in Transformer Models

"Large Transformer models have achieved impressive performance in many natural language tasks. In particular, Transformer based language models have been shown to have great capabilities in encoding factual knowledge in their vast amount of parameters. While the tasks of improving the memorization and generalization of Transformers have been widely studied, it is not well known how to make transformers forget specific old facts and memorize new ones. In this paper, we propose a new task of *explicitly modifying specific factual knowledge in Transformer models while ensuring the model performance does not degrade on the unmodified facts*. This task is useful in many scenarios, such as updating stale knowledge, protecting privacy, and eliminating unintended biases stored in the models. We benchmarked several approaches that provide natural baseline performances on this task. This leads to the discovery of key components of a Transformer model that are especially effective for knowledge modifications. The work also provides insights into the role that different training phases (such as pretraining and fine-tuning) play towards memorization and knowledge modification."

Geometry of factual knowledge and representational similarity

Work from 2024-2025 argues that whether an edit propagates is not well predicted by raw closeness alone. What seems to matter more is the **kind of geometry** the model has for facts: whether related facts lie on shared structured manifolds, support a bilinear relation-like organization, and stay aligned with the model's normal residual-stream trajectories after the edit. (3 sources)

A useful way to frame the user's question is to separate **distance** from **structure**. Recent work does suggest that factual knowledge in decoder-only models has geometric organization, but it does **not** support a simple rule like "if two facts are close in representation space, editing one will update the other." Instead, the evidence points to a more conditional story: edits propagate when the two facts are embedded in a shared and well-structured representational system, and they fail when the edit creates an isolated patch that does not integrate with that system. This is clearest in three recent lines of analysis.

First, Nishi et al. argue that related entities are embedded on internal **manifolds**, and that knowledge editing can deform these manifolds rather than cleanly rewrite one point. Their "representation shattering" hypothesis says that ROME- and MEMIT-style local weight updates can distort a shared subregion of the manifold, especially when the desired new fact is far from the old fact. On this view, geometric distance matters, but mainly as a predictor of **how much distortion** the edit causes, not as a direct predictor that nearby facts will update together in a coherent way. The implication is that nearby facts may be affected because they share compressed representational subspaces, but this effect can look like collateral damage rather than correct logical propagation. (Nishi et al., 2024)

Second, Kim et al. give a stronger and more positive geometric account of successful propagation. They report a "mechanistic link" between **representational geometry and editing consistency**: models with a **bilinear structure** are better able to propagate edits to logically related facts, while models without that

structure often fail to generalize even when the direct edit succeeds. This is a much sharper answer to the user's question than plain similarity. Their claim is not merely that related facts are close, but that facts and relations are organized in a form that supports systematic composition. Under that condition, editing one fact can carry through to entailed facts; without it, the model may retrieve the edited statement but still fail on related consequences. So the best current evidence points toward **structured relational geometry**, not raw cosine similarity or Euclidean nearness, as the more useful predictor of propagation. [\(Kim et al., 2025\)](#)

Third, Jeong et al. analyze the **residual stream trajectories** of edited knowledge and show why many edits do not integrate semantically. They find that under baseline editing methods such as ROME, the new knowledge often appears as an **isolated residual stream** that diverges from the trajectory of the model's pre-existing reference knowledge. In their interpretation, the edit drives direct activation of the target token and can bypass the model's normal reasoning path. This gives a concrete geometric picture of failed propagation: even if the edited representation reaches the right answer token, it may travel along a path that is **misaligned** with the rest of the model's factual machinery, so related one-hop or multi-hop consequences do not update with it. They also report that when editing is modified to improve semantic integration, the edited trajectories align more closely with reference knowledge, which supports the idea that **residual-stream alignment** is a better sign of coherent propagation than local answer success alone. [\(Jeong et al., 2025\)](#)

Taken together, these papers suggest a fairly clear synthesis for decoder-only models like GPT-2, GPT-J, and likely LLaMA-family systems. **Geometric proximity by itself is not a reliable predictor** that editing one fact will propagate to another. If two facts are nearby because they occupy an entangled manifold region, editing one may indeed affect the other, but that can produce interference or shattering rather than the desired update. If two facts participate in a more systematic bilinear or relation-structured geometry, propagation is more likely to be logically consistent. And if the post-edit residual stream stays isolated from the model's ordinary knowledge trajectories, then even a successful direct edit is unlikely to generalize through reasoning chains. In short, the literature increasingly favors **geometry-with-mechanism**—manifold organization, bilinear structure, and residual-stream alignment—over a simple nearest-neighbor view of factual editing outcomes. [\(Nishi et al., 2024\)](#) [\(Kim et al., 2025\)](#) [\(Jeong et al., 2025\)](#)

For the review's main question, that means the current answer is **partly yes, but not in the naive sense**. Some geometric properties of internal representations do predict whether an edit will transfer, but the strongest evidence is about **shared structured geometry and trajectory alignment**, not about bare distance or similarity alone. That helps explain why locate-and-edit methods can succeed on the edited fact yet still fail to update semantically related facts: they may write a new local retrieval path without truly reinserting the fact into the model's broader relational geometry. [\(Kim et al., 2025\)](#) [\(Jeong et al., 2025\)](#)

Evidence

(Nishi et al., 2024)

[Representation Shattering in Transformers: A Synthetic Study with Knowledge Editing](#)

"Hypothesis 3.3 (Representation shattering). Language models embed related entities on a manifold in their internal representations. KE methods distort this manifold in order to insert new facts or alter old ones, i.e., they shatter model representations. The extent of representation shattering increases with the distance between the old fact and the desired new fact on the manifold"

".(4) KE methods (ROME, MEMIT, etc.) enact local weight updates that deform these manifolds, causing representation shattering for unedited facts residing in shared sub-regions. Points (1)-(4) together lead us to believe that the fragility of KE arises from the entangled, compressed nature of factual storage, rather than just from the task of knowledge retention in and of itself."

(Kim et al., 2025)

[Bilinear representation mitigates reversal curse and enables consistent model editing](#)

"We establish a mechanistic link between representational geometry and editing consistency. Models possessing a bilinear structure successfully propagate edits to logically related facts, whereas those lacking it fail to generalize, even when the direct edit is successful (Figure 5)"

"Crucially, we established a strong link between the presence of this bilinear representation and the model's ability to both overcome the reversal curse and perform logically consistent model editing. When a fact was edited, models with this structure successfully propagated the change to entailed facts, whereas models without it failed to generalize. Our findings highlight that the path toward more reliable and editable language models lies not just in better algorithms but in shaping the fundamental geometry of their learned knowledge representations."

(Jeong et al., 2025)

STEAM: A Semantic-Level Knowledge Editing Framework for Large Language Models

"Our analysis reveals that edited knowledge is often encoded as isolated residual streams -sequences of hidden states propagated through the transformer layers which are distinct from model's pre-existing knowledge (§ 3.2). Building on this finding, additional interpretability-based analysis reveals that these residual streams reflect a direct activation for the generation of the target token o^ , bypassing the model's natural reasoning process (§ 3.3). This behavior exposes a fundamental shortcoming of current editing strategies and underscores the need for approaches that promote semantic-level integration for coherent and reliable knowledge editing"*

"To complement the qualitative findings presented in Section 3.2, we provide additional visualization results for both GPT-J and GPT-2 XL (Radford et al., 2019). These visualizations illustrate how edited knowledge propagates through the residual stream and how it aligns (or diverges) from reference knowledge across transformer layers"

"Under baseline editing methods (ROME), the edited knowledge forms a semantically isolated stream in the latent space, following a path that deviates from reference knowledge. In contrast, under STEAM-enhanced editing, the edited trajectories are more closely aligned with reference knowledge, indicating improved semantic integration."

Do internal representation similarity or localization predict edit success or propagation?

*Current evidence says ****no simple predictor works****. Neither causal localization alone nor raw representational similarity is enough to tell us whether editing one fact will cleanly transfer to another; what matters more is whether the edit captures the right subject-and-relation computation and avoids overfitting a narrow retrieval path. (9 sources)*

The strongest near-direct answer from the editing literature is that **localization and editability come apart**. Hase et al. test the central assumption behind locate-and-edit methods and find that conclusions from causal tracing "do not provide any insight into which model MLP layer would be best to edit" for changing a fact. They argue that causal tracing tells us **where factual information is carried during a forward pass**, but that this is a different question from **where to intervene to change the model's expressed fact**. So even if two facts activate nearby layers or states, that by itself does not predict that editing one there will propagate to the other. (Hase et al., 2023) This point is repeated in later editing work that explicitly notes that "the localization found by causal tracing is not predictive of edit success." (Chintam et al., 2023)

A related lesson is that many edit failures come from **editing the wrong feature**, not just the wrong layer. Recent work argues that standard subject-centered interpretations of factual recall are incomplete because they neglect **relation information**. Liu et al. say that existing interpretations mainly focus on subject knowledge and are "seriously flawed," which leads to **over-generalizing** edits; they propose a relation-focused view of recall to avoid this problem. (Liu et al., 2024) A 2025 follow-up gives a more mechanistic version of the same claim for ROME: during optimization, the edit can rely too much on the hidden state at the **last subject token** and too little on the **last relation token**, creating a kind of shortcut learning. In their account, this subject-feature bias is a main reason locate-then-edit methods become uncontrollable and alter unrelated facts linked to the same subject. (Liu et al., 2025) (Meng et al., 2022) This is highly relevant to the user's question: if two facts share subject-side representations, high internal similarity may actually predict **spurious co-editing** or over-propagation, rather than the desired update of a logically related fact.

This helps explain why simple similarity measures have limited value as predictors. If two facts are close because they share **entity/subject features**, then editing one may spread too broadly across other subject-linked associations. If two facts are related mainly through **relation structure** or reasoning chains, then subject-side closeness may miss the facts that should change. In that sense, the practical predictor is not bare geometric nearness, but whether the localized edit targets the **same causal computation** that the model

uses to combine subject and relation information for downstream answers. The recent relation-focused papers suggest that many current methods do not do this reliably. ([Liu et al., 2024](#)) ([Liu et al., 2025](#))

The broader evaluation literature reaches a similar conclusion from behavior. Later surveys and benchmarks note that locate-then-edit methods often produce **high-confidence but narrow** changes that do not generalize to related queries, diverse inputs, or multi-step reasoning. Chen et al. summarize this as a general limitation of both localization-based editors and alternatives like patching: optimizing a single knowledge item can overfit, so the edit succeeds on the target prompt but fails on related concepts and complex questions. ([Chen et al., 2025](#)) ([Zhang et al., 2024](#)) ([Zhong et al., 2023](#)) They also point to newer benchmarks that expand ripple-effect evaluation beyond narrow domains, again implying that edit propagation depends on more than a local match in representation space. ([Chen et al., 2025](#)) ([Chen et al., 2025](#))

So the best synthesis for decoder-only models is: **localization is useful, but not sufficient; similarity is suggestive, but not predictive on its own**. Causal tracing can reveal where information flows, but that does not reliably identify the best edit site. Representational overlap can signal risk of interference, especially around shared subject features, but it does not by itself tell us whether an edit will propagate correctly to a second fact. The current literature points instead to a richer predictor: whether the two facts share the **right subject-relation mechanism**, whether the edit preserves that mechanism rather than shortcutting through the subject alone, and whether the resulting change supports the model's normal inference path instead of a narrow memorized rewrite. ([Hase et al., 2023](#)) ([Chintam et al., 2023](#)) ([Liu et al., 2024](#)) ([Liu et al., 2025](#)) ([Chen et al., 2025](#))

Evidence

(Hase et al., 2023)

[Does Localization Inform Editing? Surprising Differences in Causality-Based Localization vs. Knowledge Editing in Language Models](#)

"Specifically, we show that localization conclusions from representation denoising (also known as Causal Tracing) do not provide any insight into which model MLP layer would be best to edit in order to override an existing stored fact with a new one"

". We interpret our results to suggest that Causal Tracing answers a different question than model editing does. That is, Causal Tracing answers a question about where factual information is carried in representations in a Transformer forward pass, and this question turns out to be a different question than the editing question of where is best to intervene in the Transformer in order to change the factual information it expresses."

(Chintam et al., 2023)

[Identifying and Adapting Transformer-Components Responsible for Gender Bias in an English Language Model](#)

"However, recent work has showed that although edits may be successful, the localization found by causal tracing is not predictive of edit success (Hase et al., 2023)."

(Liu et al., 2024)

[Relation Also Knows: Rethinking the Recall and Editing of Factual Associations in Auto-Regressive Transformer Language Models](#)

"Most editing works achieve knowledge editing under the guidance of existing interpretations of knowledge recall that mainly focus on subject knowledge. However, these interpretations are seriously flawed, neglecting relation information and leading to the over-generalizing problem for editing. In this work, we discover a novel relation-focused perspective to interpret the knowledge recall of transformer LMs during inference and apply it on single knowledge editing to avoid over-generalizing."

(Liu et al., 2025)

[Causal Path Alignment: Anchoring the Optimization Trajectory for Controllable In-Parameter Knowledge Editing](#)

"A prominent line of previous works achieve knowledge editing via localizing the decisive parameters during the recall of factual associations of auto-regressive transformer language models and then modify the located parameters with regularization constrained updates, known as the locatethen-edit methods [6,(Meng et al., 2022)8,9]. However, these methods are uncontrollable, frequently resulting in unexpected alterations to most unrelated factual associations linked to the subject of the edited factual association. [10,11]"

"In this work, we unveil that this failure of locate-then-edit methods stems from a shortcut learning [12] issue during the optimization process of editing, as exemplified by the investigation on the representative ROME editing (Meng et al., 2022). Specifically, through the analysis of the significance of gradients at different token positions during the optimization of ROME, we discover that the hidden representations traversing the layers at the last subject token position and the last relation token position are two crucial features that models learn during the optimization process. However, further exploration into the contribution of these two features to the predictions of factual associations in edited models shows that ROME editing exhibits a bias towards over-learning the feature at the last subject position while inadequately learning the feature at the last relation position. Since subject-related attributes are mainly recalled at the last subject token position (Meng et al., 2022) and relation-related attributes are mainly recalled at the last relation token position [10] (i.e. typically the last token position), the two features can be regarded as the subject feature and the relation feature respectively. In conclusion, ROME editing falls into the trap of shortcut learning of the subject feature while neglecting the relation feature, resulting in uncontrollable editing (Fig. 1)."

(Meng et al., 2022)

Locating and Editing Factual Associations in GPT

"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at <https://rome.baulab.info/>"

(Chen et al., 2025)

An information-theoretic framework for robust large language model editing

"Current model editing techniques frequently struggle to generalize corrections beyond narrow domains, leading to unintended consequences and limiting their practical impact"

"Contemporary approaches typically focus on identifying and directly modifying localized, knowledge-sensitive regions in the network, a process termed "locate-then-edit" (L&E) (Meng et al., 2022)- (Fang et al., 2024). Alternatively, methods such as transformer patching (Huang et al., 2023) bypass localization by optimizing a new neuron for each edit. Despite their promise, both paradigms face notable limitations: optimizing a single piece of knowledge can often result in overfitting, leading to updates that are highly confident yet narrow and may fail to generalize to related queries or concepts (Zhang et al., 2024). As a consequence, performance on complex questions, especially those requiring multi-step reasoning or the handling of diverse inputs, is compromised (Zhong et al., 2023), (Chen et al., 2025)."

(Zhang et al., 2024)

Uncovering Overfitting in Large Language Model Editing

"Knowledge editing has been proposed as an effective method for updating and correcting the internal knowledge of Large Language Models (LLMs). However, existing editing methods often struggle with complex tasks, such as multi-hop reasoning. In this paper, we identify and investigate the phenomenon of Editing Overfit, where edited models assign disproportionately high probabilities to the edit target, hindering the generalization of new knowledge in complex scenarios. We attribute this issue to the current editing paradigm, which places excessive emphasis on the direct correspondence between the input prompt and the edit target for each edit sample. To further explore this issue, we introduce a new benchmark, EVOKE (EValuation of Editing Overfit in Knowledge Editing), along with fine-grained evaluation metrics. Through comprehensive experiments and analysis, we demonstrate that Editing Overfit is prevalent in current editing methods and that common overfitting mitigation strategies are ineffective in knowledge editing. To overcome this, inspired by LLMs' knowledge recall mechanisms, we propose a new plug-and-play strategy called Learn the Inference (LTI), which introduce a Multi-stage Inference Constraint module to guide the edited models in recalling new knowledge similarly to how unedited LLMs leverage knowledge through in-context learning. Extensive experimental results across a wide range of tasks validate the effectiveness of LTI in mitigating Editing Overfit."

(Zhong et al., 2023)

MQuAKE: Assessing Knowledge Editing in Language Models via Multi-Hop Questions

"Current evaluation paradigms are extremely limited, mainly validating the recall of edited facts, but changing one fact should cause rippling changes to the model's related beliefs. If we edit the UK Prime Minister to now be Rishi Sunak, then we should get a different answer to Who is married to the British Prime Minister? In this work, we present a benchmark, MQuAKE (Multi-hop Question Answering for Knowledge Editing), comprising multi-hop questions that assess whether edited models correctly answer questions

where the answer should change as an entailed consequence of edited facts. While we find that current knowledge-editing approaches can recall edited facts accurately, they fail catastrophically on the constructed multi-hop questions"

".• We evaluate state-of-the-art knowledge-editing methods on MQUAKE, testing from editing facts mentioned in one question to editing facts mentioned in a large set of questions. The latter setting evaluates approaches that are designed to handle many edits, such as MEMIT (Meng et al., 2022b). Surprisingly, existing knowledge-editing methods often perform well on answering questions that are paraphrases of the edited fact but fail drastically on questions where the answer should change as a consequence of an edited fact. For example, a GPT-J model edited by ROME (Meng et al., 2022)) can only answer 7.6% of multi-hop questions in MQUAKE-CF, even though it could answer 43.4% of the questions before editing"

".• Fine-tuning (FT) simply performs gradient descent on the edits to update model parameters. We follow (Zhu et al., 2020) and fine-tune one layer in the model with a norm constraint on weight changes"

".• MEND (Mitchell et al., 2021) trains a hypernetwork to produce weight updates by transforming the raw fine-tuning gradients given an edited fact"

".• ROME (Meng et al., 2022) first localizes the factual knowledge at a certain layer in the Transformer architecture, and then updates the feedforward network in that layer to insert the new facts"

".• MEMIT (Meng et al., 2022b)"

(Chen et al._1, 2025)

UniEdit: A Unified Knowledge Editing Benchmark for Large Language Models

"Model editing aims to enhance the accuracy and reliability of large language models (LLMs) by efficiently adjusting their internal parameters. Currently, most LLM editing datasets are confined to narrow knowledge domains and cover a limited range of editing evaluation. They often overlook the broad scope of editing demands and the diversity of ripple effects resulting from edits. In this context, we introduce UniEdit, a unified benchmark for LLM editing grounded in open-domain knowledge. First, we construct editing samples by selecting entities from 25 common domains across five major categories, utilizing the extensive triple knowledge available in open-domain knowledge graphs to ensure comprehensive coverage of the knowledge domains. To address the issues of generality and locality in editing, we design an Neighborhood Multi-hop Chain Sampling (NMCS) algorithm to sample subgraphs based on a given knowledge piece to entail comprehensive ripple effects to evaluate. Finally, we employ proprietary LLMs to convert the sampled knowledge subgraphs into natural language text, guaranteeing grammatical accuracy and syntactical diversity. Extensive statistical analysis confirms the scale, comprehensiveness, and diversity of our UniEdit benchmark. We conduct comprehensive experiments across multiple LLMs and editors, analyzing their performance to highlight strengths and weaknesses in editing across open knowledge domains and various evaluation criteria, thereby offering valuable insights for future research endeavors."

Causal localization and intervention methods

Causal localization tools ask not just where information is present, but which internal states actually matter for producing a factual answer. In decoder-only models, these methods are useful for finding candidate edit sites and circuits, but they do not by themselves predict whether an edit will generalize or propagate to related facts. (4 sources)

Mechanistic work on decoder-only LMs has increasingly used **causal interventions** to move beyond correlational probes. The notable starting point for factual editing is **causal tracing** in ROME: Meng et al. corrupt selected hidden states during a forward pass, then restore individual activations to test which ones are decisive for the model's factual prediction. Their main finding is that factual recall depends strongly on a set of **middle-layer feed-forward computations while processing subject tokens**, which motivated editing those modules directly with a rank-one update. This is the key localization story behind locate-and-edit methods in GPT-style models. ([Meng et al., 2022](#))

A broader family of intervention tools extends this logic from "important activations" to **circuits and paths**. In mechanistic interpretability, **activation patching** and closely related **path patching** replace activations from one run into another to test whether a component or route is causally needed for a behavior. A notable example is the GPT-2 small indirect-object-identification analysis, which used causal interventions to identify a multi-head circuit and evaluate it by faithfulness, completeness, and minimality. This matters here because it shows how researchers can trace **signal propagation across attention heads and MLPs**, rather than only naming a single "knowledge layer." ([Wang et al., 2022](#)) ([Palikhe et al., 2025](#))

A related intervention style is **interchange intervention**, where one swaps internal states between examples

to test whether a hypothesized variable or component carries the relevant computation. In recent factual-recall work beyond transformers, Sharma et al. apply **causal tracing** or **interchange interventions** to localize the components in Mamba responsible for recalling facts, and they find a pattern that closely matches autoregressive transformers: strong causal effects appear in **middle layers at the last subject token**, while interventions in later layers matter most at the **last token of the prompt**. They then show that rank-one edits can insert facts at those locations, which strengthens the general methodological point that causal localization can identify reusable edit-relevant sites across autoregressive architectures. ([Sharma et al., 2024](#))

For the present review, the main use of these methods is conceptual. They help separate at least three questions that are easy to conflate: **where fact-related information is represented**, **which components causally mediate its recall**, and **which parameter update site will best change the model's behavior**. Causal tracing, activation patching, path patching, and interchange interventions are strong tools for the first two questions, and they have been essential for building mechanistic hypotheses behind ROME-like editors. But, as the previous section emphasized, they are not enough to answer the third question, much less to predict ripple effects or multi-hop propagation after the edit. ([Meng et al., 2022](#)) ([Wang et al., 2022](#)) ([Sharma et al., 2024](#))

This is why later discussions treat causal localization as **necessary but not sufficient**. Palikhe et al. summarize ROME as showing that some facts sit in editable subnetworks, but they also stress the limits of this style of analysis: it assumes fairly fixed activation paths, is strongest for factual recall rather than abstract or multi-hop reasoning, and related circuit tools such as path patching often require full white-box access and substantial compute. In practice, this means localization can reveal where a model retrieves a fact, yet still miss the distributed computations needed for composition, contradiction handling, or consequence propagation. ([Palikhe et al., 2025](#)) ([Wang et al., 2022](#)) ([Meng et al., 2022](#))

The synthesis for the user's question is therefore fairly sharp. **Causal localization methods are among the best tools we have for finding where an edited fact is routed through a decoder-only LM, but they do not show that geometric closeness between two facts will make an edit propagate from one to the other.** What they can show is whether the two facts seem to rely on overlapping components, tokens, layers, or paths. That overlap is useful evidence about possible interference or transfer, but predicting successful propagation still requires the extra ingredients discussed earlier: the right subject-relation mechanism, structured geometry, and alignment with the model's normal inference trajectory.

Evidence

(Meng et al., 2022)

[Locating and Editing Factual Associations in GPT](#)

"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at <https://rome.baulab.info/>"

(Wang et al., 2022)

[Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 small](#)

"Research in mechanistic interpretability seeks to explain behaviors of machine learning models in terms of their internal components. However, most previous work either focuses on simple behaviors in small models, or describes complicated behaviors

in larger models with broad strokes. In this work, we bridge this gap by presenting an explanation for how GPT-2 small performs a natural language task called indirect object identification (IOI). Our explanation encompasses 26 attention heads grouped into 7 main classes, which we discovered using a combination of interpretability approaches relying on causal interventions. To our knowledge, this investigation is the largest end-to-end attempt at reverse-engineering a natural behavior "in the wild" in a language model. We evaluate the reliability of our explanation using three quantitative criteria—faithfulness, completeness and minimality. Though these criteria support our explanation, they also point to remaining gaps in our understanding. Our work provides evidence that a mechanistic understanding of large ML models is feasible, opening opportunities to scale our understanding to both larger models and more complex tasks."

(Palikhe et al., 2025)

[Towards Transparent AI: A Survey on Explainable Large Language Models](#)

"ROME (Rank-One Model Editing) (Meng et al., 2022), offers a localized editing framework to modify factual knowledge in GPT-2 (Radford et al., 2019) by targeting specific feedforward layer neurons, showing some facts are stored in editable subnetworks (e.g., changing "Paris is the capital of France" to "Florida is the capital of France"). This links outputs to identifiable components, enhancing explainability and enabling direct edits. However, ROME is limited to factual recall, assumes fixed activation paths, and struggles with abstract or multi-hop reasoning requiring complex interactions"

".Related efforts, such as Path Patching (Wang et al., 2022) and Transformer Circuits [50], analyze attention and MLP contributions to trace signal propagation, uncovering reusable, modular components. However, these methods demand full whitebox access and are computationally intensive, hindering scalability."

(Sharma et al., 2024)

[Locating and Editing Factual Associations in Mamba](#)

"we apply causal tracing or interchange interventions to localize key components inside Mamba that are responsible for recalling facts, revealing that specific components within middle layers show strong causal effects at the last token of the subject, while the causal effect of intervening on later layers is most pronounced at the last token of the prompt, matching previous findings on autoregressive transformers. Second, we show that rank-one model editing methods can successfully insert facts at specific locations, again resembling findings on transformer LMs. Third, we examine the linearity of Mamba's representations of factual relations."

Compositional and multi-hop reasoning failures after editing

After editing, decoder-only LMs often know the new one-hop fact but still fail when they must use it inside a reasoning chain. Recent mechanistic work suggests this is not just a benchmark gap: single-fact editors often change shallow recall pathways, while multi-hop use depends on deeper, sequential computations that involve implicit subjects and later-layer neuron dynamics. (5 sources)

A useful synthesis of the post-RippleEdits/MQuAKE literature is that **multi-hop failure looks mechanistically different from ordinary edit failure**. The problem is not only that ROME, MEMIT, or MEND sometimes miss the target fact; it is that they often change the model where **single-hop** retrieval happens, without fully changing the later computations that support chained use of that fact in downstream reasoning. Zhang et al. make this point directly: methods built from single-hop prompts update "shallow MLP layers" but often fail to propagate the change into "deeper layers," so old knowledge can still reappear when a question triggers implicit multi-hop recall rather than direct factual completion ([Zhang et al., 2, 2024](#)) ([Meng et al., 2022](#)) ([Meng et al., 1, 2022](#)) ([Mitchell et al., 2021](#)). In their account, the retrieval pattern for multi-hop questions differs from the one emphasized by ROME-style causal tracing: in single-hop recall, the explicit subject can retrieve associated knowledge from earlier or middle MLP layers, but in multi-hop settings an **implicit subject** emerges inside the reasoning process and guides answer formation by pulling relevant knowledge from **later MLP layers** ([Zhang et al., 2, 2024](#)) ([Meng et al., 2022](#)). That gives a concrete reason why editing one fact may not propagate to another even when the two are related: the edited association may be reachable through the direct prompt, yet the model may still use untouched deeper-layer machinery when it has to compose that fact with another step.

The 2025 neuron-level analysis by Yang et al. pushes this into a sharper mechanistic story. They argue that multi-hop factual recall depends on **query-value pathways** in which implicit subjects act like **query neurons** that successively activate **value neurons** across layers, accumulating information toward the final answer. Their claim is that prior knowledge editing methods largely ignored this dynamic, which is why performance

drops are especially severe when the edited fact must be accessed through an intermediate hidden entity rather than through the original edited subject string (Yang et al., 2025). This account fits the behavioral pattern seen earlier in MQUAKE-style failures: direct recall can look fixed, but the model has not actually updated the internal chain that reconstructs the answer under composition. On this view, what breaks is not only “knowledge storage,” but **knowledge routing during reasoning**. Editing one endpoint in the chain does little if the intermediate query neurons still call the old value pathway.

This deeper-layer picture also helps refine the answer to the user’s geometry question. If propagation depended mainly on representational closeness between two facts, then many multi-hop consequences should update once the base fact is edited. But the mechanistic evidence here points elsewhere: **whether a second fact changes depends on whether the edit reaches the later-layer causal path that the model uses when an implicit subject appears during reasoning**, not just on whether the two facts are nearby in activation space (Zhang et al., 2024) (Yang et al., 2025). In practice, closeness may correlate with some interference, but the stronger predictor of compositional transfer seems to be overlap in these sequential query-value or implicit-subject pathways. That is why newer methods aimed at multi-hop editing try to localize and modify neuron-level attributions along the chain itself, rather than only rewriting the earlier subject-to-value map used in one-hop prompts (Yang et al., 2025).

So the main synthesis is: **post-edit multi-hop failure is best explained as a mismatch between where current editors write and where compositional reasoning actually runs**. ROME and MEMIT remain notable because they showed that factual associations in decoder-only models can be edited through localized MLP updates (Meng et al., 2022) (Meng et al., 2022), and MEND showed that large models can be edited efficiently with learned update rules (Mitchell et al., 2021). But the newer mechanistic work suggests that these edits often stop short of the deeper-layer dynamics needed for belief revision under composition. This is why one edited fact can remain isolated, why old answers can return in multi-step settings, and why internal similarity alone is a weak predictor of whether the edit will propagate through reasoning chains (Zhang et al., 2024) (Yang et al., 2025).

Evidence

(Zhang et al., 2024)

Locate-then-edit for Multi-hop Factual Recall under Knowledge Editing

"Previous methods leveraging single-hop prompts for editing are insufficient as they only update the relevant knowledge in the shallow MLP layers but fail to propagate the changes to deeper layers. The model retains some of the old single-hop knowledge that only activated by additional implicit multi-hop fact recall mechanisms"

"our results indicate that in the multi-hop scenario, the implicit subject guides the emergence of the final answer by retrieving relevant knowledge from the later MLP layers. This contrasts sharply with the single-hop cases (Meng et al., 2022)2023), where the subject information is used to retrieve information from earlier MLP layers"

"An example is ROME (Meng et al., 2022), which explored the knowledge storage mechanisms in single-hop factual recall tasks based on causal tracing methods and proposed the Rank-One Model Editing method. Together with KN (Dai et al., 2021), it pioneered a paradigm of locate-then-edit, providing guidance for subsequent editing methods. The later extended versions, MEMIT (Meng et al., 2022), MALMEN (Tan et al., 2023), and EMMET (Gupta et al., 2024), further improved ROME by addressing its limitations in large-scale editing, enabling comprehensive edits in a single operation while demonstrating exceptional performance"

"Additionally, MEND (Mitchell et al., 2021) trained a hypernetwork to efficiently predict LLM weight updates, enabling rapid knowledge editing"

"However, these editing methods primarily focus on knowledge updates in specific layers and lack in-depth optimization for knowledge integration and application in multi-hop reasoning, rendering them inadequate for multi-hop questions."

(Meng et al., 2022)

Locating and Editing Factual Associations in GPT

"We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying

neuron activations that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward weights to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods. To perform a more sensitive evaluation, we also evaluate ROME on a new dataset of counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at <https://rome.baulab.info/>"

(Meng et al., 2022)

Mass-Editing Memory in a Transformer

"De (Cao et al., 2021) proposes a Knowledge Editor (KE) hypernetwork that edits a standard model by predicting updates conditioned on new factual statements. In a study of KE, Hase et al. (2021) find that it fails to scale beyond a few edits, and they scale an improved objective to 10 beliefs. MEND (Mitchell et al., 2021) also adopts meta-learning, inferring weight updates from the gradient of the inserted fact. To scale their method, (Mitchell et al., 2022) proposes SERAC, a system that routes rewritten facts through a different set of parameters while keeping the original weights unmodified; they demonstrate scaling up to 75 edits. Rather than meta-learning, our method employs direct parameter updates based on an explicitly computed mapping"

"Our method is closely related to (Meng et al., 2022), which also updates GPT as an explicit associative memory. Unlike the single-edit approach taken in that work, we modify a sequence of layers and develop a way for thousands of modifications to be performed simultaneously"

"MEMIT targets the weights of transformer modules that we determine to be causal mediators of factual knowledge recall"

"In practical settings, we may wish to MEMIT modifies transformer weights to edit memories, e.g., "Michael Jordan now plays the sport baseball," while (c) maintaining generalization, specificity, and fluency at scales beyond other methods. As Section 5.2.2 details, editing score is the harmonic mean of efficacy, generalization, and specificity metrics."

(Mitchell et al., 2021)

Fast Model Editing at Scale

"While large pre-trained models have enabled impressive results on a variety of downstream tasks, the largest existing models still make errors, and even accurate predictions may become outdated over time. Because detecting all such failures at training time is impossible, enabling both developers and end users of such models to correct inaccurate outputs while leaving the model otherwise intact is desirable. However, the distributed, black-box nature of the representations learned by large neural networks makes producing such targeted edits difficult. If presented with only a single problematic input and new desired output, fine-tuning approaches tend to overfit; other editing algorithms are either computationally infeasible or simply ineffective when applied to very large models. To enable easy post-hoc editing at scale, we propose Model Editor Networks using Gradient Decomposition (MEND), a collection of small auxiliary editing networks that use a single desired input-output pair to make fast, local edits to a pre-trained model's behavior. MEND learns to transform the gradient obtained by standard fine-tuning, using a low-rank decomposition of the gradient to make the parameterization of this transformation tractable. MEND can be trained on a single GPU in less than a day even for 10 billion+ parameter models; once trained MEND enables rapid application of new edits to the pre-trained model. Our experiments with T5, GPT, BERT, and BART models show that MEND is the only approach to model editing that effectively edits the behavior of models with more than 10 billion parameters. Code and data available at <https://sites.google.com/view/mend-editing>."

(Yang et al., 2025)

ACE: Attribution-Controlled Knowledge Editing for Multi-hop Factual Recall

"Large Language Models (LLMs) require efficient knowledge editing (KE) to update factual information, yet existing methods exhibit significant performance decay in multi-hop factual recall. This failure is particularly acute when edits involve intermediate implicit subjects within reasoning chains. Through causal analysis, we reveal that this limitation stems from an oversight of how chained knowledge is dynamically represented and utilized at the neuron level. We discover that during multi hop reasoning, implicit subjects function as query neurons, which sequentially activate corresponding value neurons across transformer layers to accumulate information toward the final answer, a dynamic prior KE work has overlooked. Guided by this insight, we propose ACE: Attribution-Controlled Knowledge Editing for Multi-hop Factual Recall, a framework that leverages neuron-level attribution to identify and edit these critical query-value (Q-V) pathways"

"Our analysis further reveals more fine-grained activation patterns in Qwen3 and demonstrates that the semantic interpretability of value neurons is orchestrated by query-driven accumulation. These findings establish a new pathway for advancing KE capabilities based on the principled understanding of internal reasoning mechanisms."